

Column j_1 below Row i is covered as follows: A Domino covers $(i+1, j_1)$ and $(i+2, j_1)$. Another Domino covers $(i+3, j_1)$ and $(i+4, j_1)$. This process is continued until a Domino covers $(7, j_1)$ and $(8, j_1)$.

Suppose both j_1 and j_2 are even.

$\therefore \exists$ integers K_1 and K_2 s.t. $j_1 = 2K_1$ and $j_2 = 2K_2$ (From Defn. 1)

$(j_1 - j_2) = 2K_1 - 2K_2 = 2(K_1 - K_2)$. By Defn. 1, $(j_1 - j_2)$ is even. $\square$

Suppose both j_1 and j_2 are odd.

From Defn. 1, $\exists$ integers K_1 and K_2 s.t. $j_1 = (2K_1 + 1)$ and $j_2 = (2K_2 + 1)$

$(j_1 - j_2) = (2K_1 + 1) - (2K_2 + 1) = 2(K_1 - K_2)$. By Defn. 1, $(j_1 - j_2)$ is even.

$\therefore$ A Domino covers Row 8, Column $(j_1 + 1)$ and Row 8, Column $(j_1 + 2)$. Another Domino covers Row 8, Column $(j_1 + 3)$ and Row 8, Column $(j_1 + 4)$. This process is continued until a Domino covers $(8, j_2 - 1)$ and $(8, j_2)$.

$\therefore (8 - j_2 - 1) = 2 \times 4 - (2(x+y) + 1) - 1 = \boxed{2(4 - x - y - 1)}$

By defn. 1, ~~Column j_2 below Row 8 is covered~~ $(8 - j_2 - 1)$ is even.

Column j_2 below Row i_2 is covered as follows: A Domino covers $(i_2 + 1, j_2)$ and $(i_2 + 2, j_2)$. Another Domino covers $(i_2 + 3, j_2)$ and $(i_2 + 4, j_2)$. This process is continued until a Domino covers $(6, j_2)$ and $(7, j_2)$.

$i_2 - 1 = 2(x+y) + 1 - 1 = 2(x+y)$. By Defn. 1, $(i_2 - 1)$ is even.

Column j_2 above Row i_2 is covered as follows: A Domino covers $(1, j_2)$ and $(2, j_2)$. Another Domino covers $(3, j_2)$ and $(4, j_2)$. This process continues until a Domino covers $(i_2 - 2, j_2)$ and $(i_2 - 1, j_2)$.

A Domino covers Row 1, Column $(j_1 + 1)$ and Row 1, Column $(j_1 + 2)$. Another Domino covers Row 1, Column $(j_1 + 3)$ and Row 1, Column $(j_1 + 4)$. This process is continued until a Domino covers $(1, j_2 - 1)$ and $(1, j_2)$.

$(i_1 - 1) - 1 = i_1 - 2 = 2y - 2 = 2(y - 1)$. By Defn. 1, $(i_1 - 2)$ is even. $\square$